

$$\Delta H_f^\circ = +69.36 \text{ kJ/mol}$$

- (a) The reaction is endothermic because it is absorbing heat. You would need to heat the reactor to keep the temperature constant. If it ran adiabatically, then the temperature would drop and cool up. More energy is required to break the bonds from the reactants than to form the product bonds. ✓

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(b) ΔU_f° ? $\Delta U_f^\circ(T) = \Delta H_f^\circ(T) - RT \left[\sum_{\text{Products}} V_i - \sum_{\text{Reactants}} V_i \right]$

$$\left[\sum_{\text{P}} V_i - \sum_{\text{R}} V_i \right] \rightarrow (1+2+5) - 0 = 7$$

$$\Delta U_f^\circ(T) = +69.36 \frac{\text{kJ}}{\text{mol}} - 8.314 \frac{\text{J}}{\text{mol}\cdot\text{K}} \times (25^\circ\text{C} + 273) \times (7)$$

$$\Delta U_f^\circ(T) = 69.36 - 0.0083 \frac{\text{kJ}}{\text{mol}} (298) (7)$$

$$\Delta U_f^\circ(T) = 52.017 \frac{\text{kJ}}{\text{mol}} \quad \left[\text{amount of energy from the reactants} \right] \rightarrow \text{Internal energy}$$

(c) $\Delta U = Q + W$ $\Delta U = Q$ $\Delta U_f n_{\text{CaC}_2} = Q$ $m = 150 \text{ g}$ $M = 64.1 \text{ g/mol}$
 ⊗ constant V

$$n_{\text{CaC}_2} = \frac{m}{M} = 2.34 \text{ mol} \quad 2.34 \text{ mol} (52.017) \frac{\text{kJ}}{\text{mol}} = Q = 121.72 \text{ kJ} \quad \text{Heat out of the reactor}$$

(Murphy 6.32)

$$U_{\text{sys},f} - U_{\text{sys},i} = Q$$

For Water
 $C_D = C_V = 4.18 \frac{\text{J}}{\text{g}\cdot^\circ\text{C}}$

$$U_{\text{sys},f} - U_{\text{sys},i} = Q = m(C_V(T_f - T_i)) \Rightarrow 650 \text{ g} \times 4.18 \frac{\text{J}}{\text{g}\cdot^\circ\text{C}} (36.2 - 23.6)^\circ\text{C} = 34234.2 \text{ J} \approx 34.234 \text{ kJ}$$

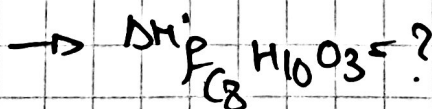
$$\Delta H_f^\circ = \Delta U_f^\circ + R(T)$$

$$n_{\text{element}} = \frac{m}{M} = \frac{2.7 \text{ g}}{154 \text{ g/mol}} = 0.0175 \text{ g/mol}$$

$$\Delta H_f^\circ = 1956.23 \frac{\text{kJ}}{\text{g}\cdot\text{mol}} + 0.008314 \frac{\text{kJ}}{\text{g}\cdot\text{mol}\cdot\text{K}} (298) \text{ K} \quad \Delta U_f n_f = Q$$

$$\frac{34.234 \text{ kJ}}{0.0175 \text{ g/mol}} = 1956.23$$

$$\Delta H_f^\circ = 1958.7 \frac{\text{kJ}}{\text{g}\cdot\text{mol}}$$



Murphy 6.34)



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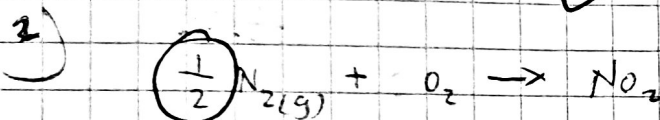
$Q = -600 \text{ kJ}$ desired $(-100 \text{ kJ/x} \times 6 \text{ x})$ [released (-)]

$$\frac{-600 \text{ kJ}}{-830.5 \text{ kJ/mol}} = 0.722 \text{ mol Fe}_2\text{O}_3 \times \frac{2 \text{ mol Fe}}{1 \text{ mol Fe}_2\text{O}_3} = 1.44 \text{ mol Fe}$$

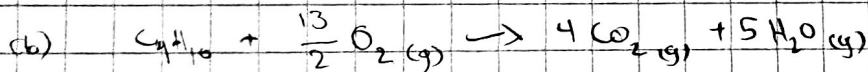
$$1.44 \text{ mol Fe} \times \frac{56 \text{ g}}{1 \text{ mol}} = \boxed{80.64 \text{ g Fe}}$$

Oxygen $0.722 \text{ mol Fe}_2\text{O}_3 \times \frac{160 \text{ g Fe}_2\text{O}_3}{1 \text{ mol Fe}_2\text{O}_3} = 115 \text{ g Fe}_2\text{O}_3$

am $115 \text{ g} - 80 \text{ g} = 35 \text{ g}$ $\% \frac{115 - 80}{80} \times 100 = 43.75\%$



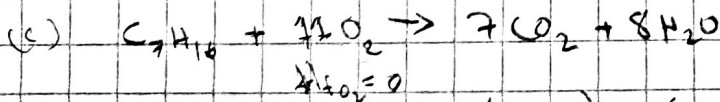
8 $\Delta H_r^\circ = H_f^\circ \text{NO}_2 - \frac{1}{2} H_f^\circ \text{N}_2 + H_f^\circ \text{O}_2$ $\Delta H_r^\circ = 33.8 \text{ kJ/mol}$ $\times 2$ 180.74



$$\Delta H_r = 4 H_f^\circ \text{CO}_2 + 5 H_f^\circ \text{H}_2\text{O} - H_f^\circ \text{C}_4\text{H}_{10} - \frac{13}{2} H_f^\circ \text{O}_2$$

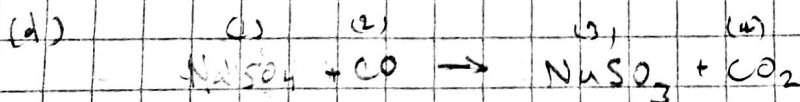
$$\Delta H_r = 4 \times (393.5) + 5(-242) - (-125) = \boxed{\Delta H_r = 489 \text{ kJ/mol}}$$

-2121.2



$$\Delta H_r = 7(393.5) + 8(-242) - (-225.2) = \boxed{\Delta H_r = 693.7 \text{ kJ/mol}}$$

-3855



$$\Delta H_r = (3) + (4) - (2) - (1)$$

$$(294.15) + (393.5) - (-99) - (-1387.1) = \boxed{\Delta H_r = 2177.75 \text{ kJ/mol}}$$

-138.2

5)

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